- Phosphine is used for producing smoke screen as it gives large smoke.
- When a slow stream of chlorine is passed over white phosphorous, phosphorous trichloride is formed.
- phosphorus trichloride: and Phosphorous pentachloride are used as a chlorinating agent
- Oxygen is paramagnetic. It exists in two allotropic forms namely dioxygen (O_2) and ozone or trioxygen (O_3).
- Ozone is commonly used for oxidation of organic compounds.
- Sulphur exists in crystalline as well as amorphous allotrophic forms. The crystalline form includes rhombic sulphur (α sulphur) and monoclinic sulphur (β sulphur). Amorphous allotropic form includes plastic sulphur (γ sulphur), milk of sulphur and colloidal sulphur.
- Sulphuric acid can be manufactured by lead chamber process, cascade process or contact process.
- When dissolved in water, it forms mono ($H_2SO_4.H_2O$) and dihydrates ($H_2SO_4.2H_2O$) and the reaction is exothermic.
- The halogens are present in combined form as they are highly reactive.
- Chlorine is manufactured by the electrolysis of brine in electrolytic process or by oxidation of HCl by air in Deacon's process.
- Chlorine is a strong oxidising and bleaching agent because of the nascent oxygen.
- When three parts of concentrated hydrochloric acid and one part of concentrated nitric acid are mixed, Aquaregia (Royal water) is obtained. This is used for dissolving gold, platinum etc...
- The hydrogen halides are extremely soluble in water due to the ionisation.
- Each halogen combines with other halogens to form a series of compounds called inter halogen compounds.
- Fluorine reacts readily with oxygen and forms difluorine oxide (F_2O) and difluorine dioxide (F_2O_2) where it has a -1 oxidation state.
- All the noble gases occur in the atmosphere.
- They are extremely stable and have a small tendency to gain or lose electrons.
- Sodium per xenate is very much known for its strong oxidizing property.
- The inertness of noble gases is an important feature of their practical uses.

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